

1. EPICs & Requirement Mapping

The Lab 1 requirements were converted into five coherent Agile epics. This preserves the functional and non-functional scope while organizing the work into deliverable themes.

Epic	Name	Lab 1 basis	Description
EPIC-1	Audit Log Ingestion	FR-001	Ingest enterprise application audit logs in structured JSON and make valid events available for processing.
EPIC-2	PII Detection & GDPR Masking	FR-002, FR-003, NFR-002	Parse audit logs, identify sensitive PII, and mask protected fields before persistence in accordance with GDPR requirements.
EPIC-3	Protected Storage & Integrity	FR-004, NFR-002	Persist protected audit records with SHA-256 integrity hashes so the audit history is tamper-evident.
EPIC-4	Audit Search & Compliance Export	FR-005	Allow authorized stakeholders to query/filter historical records and generate immutable, verifiable audit exports.
EPIC-5	High-Performance & Secure Audit Access	NFR-001, NFR-002	Provide secure, compliant audit access with complex filtering across large historical datasets within the required performance target.

2. Agile Backlog & Story Point Assignments

The backlog contains 10 user stories grouped under the five Jira epics. Priorities are set to High and estimates use the Fibonacci sequence, as required by the handout.

Issue	Story	Epic	Priority	SP
KAN-6	US-1.1 — Ingest JSON Audit Logs	Audit Log Ingestion	High	5
KAN-7	US-1.2 — Validate Audit Input	Audit Log Ingestion	High	3
KAN-8	US-2.1 — Detect Sensitive PII	PII Detection & GDPR Masking	High	5
KAN-9	US-2.2 — Mask PII Before Storage	PII Detection & GDPR Masking	High	5
KAN-10	US-3.1 — Persist Protected Records	Protected Storage & Integrity	High	3
KAN-11	US-3.2 — Generate SHA-256 Integrity Hash	Protected Storage & Integrity	High	5
KAN-12	US-4.1 — Query & Filter Audit History	Audit Search & Compliance Export	High	5
KAN-13	US-4.2 — Generate Immutable Audit Export	Audit Search & Compliance Export	High	8
KAN-14	US-5.1 — Secure Authorized Audit Access	High-Performance & Secure Audit Access	High	3
KAN-15	US-5.2 — Meet Large-Scale Query Target	High-Performance & Secure Audit Access	High	8

Total backlog estimate: 50 story points. Planning-poker rationale: 3-point stories represent contained work; 5-point stories represent moderate complexity/integration; 8-point stories carry higher uncertainty, validation, security, or performance risk.

3. Jira Backlog Evidence

The following screenshot is the captured Jira backlog showing all 10 work items, their Jira issue keys, and their parent epics.

SafariFileEditViewHistoryBookmarksWindowHelp

satya-kallakuri.atlassian.net

Agent Sprint Simulation - Backlog - Jira

Create Final Deliverable

Jira

Search

Create

Upgrade

🔔🔍⚙️👤

👤 For you

🕒 Recent

★ Starred

🔍 Ap...

📅 Plans

📍 Spaces

Recent

🔍 Agent Sprint Simulation

🔍 My discovery project

📅 More spaces

🔍 Filters

📅 Dashboards

👤 Teams

🎯 Goals

🔍 Projects

⚙️ Customize sidebar

Spaces

🔍 Agent Sprint Simulation

🌐 Summary📅 List📊 Board🔧 Development📄 Forms📅 Timeline📄 Docs📅 Backlog+

🔍 Search backlog👤📄 Filter

📊 Board (10 work items)

1000

+

📌 KAN-6Ingest JSON Audit Logs

Audit Log Ingestion

To Do

👤

📌 KAN-7Validate Audit Input

Audit Log Ingestion

To Do

👤

📌 KAN-8Detect Sensitive PII

PII Detection & G...

To Do

👤

📌 KAN-9Mask PII Before Storage

PII Detection & G...

To Do

👤

📌 KAN-10Persist Protected Records

Protected Storage ...

To Do

👤

📌 KAN-11Generate SHA-256 Integrity Hash

Protected Storage ...

To Do

👤

📌 KAN-12Query & Filter Audit History

Audit Search & Co...

To Do

👤

📌 KAN-13Generate Immutable Audit Export

Audit Search & Co...

To Do

👤

📌 KAN-14Secure Authorized Audit Access

High-Performance...

To Do

👤

📌 KAN-15Meet Large-Scale Query Target

High-Performance...

To Do

👤

+ Create

🔍 Quickstart

Figure 1. Jira backlog with the 10 created work items and epic associations.

4. Sprint 1 Simulation

Planned duration: 1 week **Stories:** KAN-6 to KAN-10 **Planned capacity:** 21 SP

Issue	Story	SP	Final status
KAN-6	Ingest JSON Audit Logs	5	Done
KAN-7	Validate Audit Input	3	Done
KAN-8	Detect Sensitive PII	5	Done
KAN-9	Mask PII Before Storage	5	Done
KAN-10	Persist Protected Records	3	Done

Captured evidence of the active sprint shows work moving from To Do into In Progress before completion.

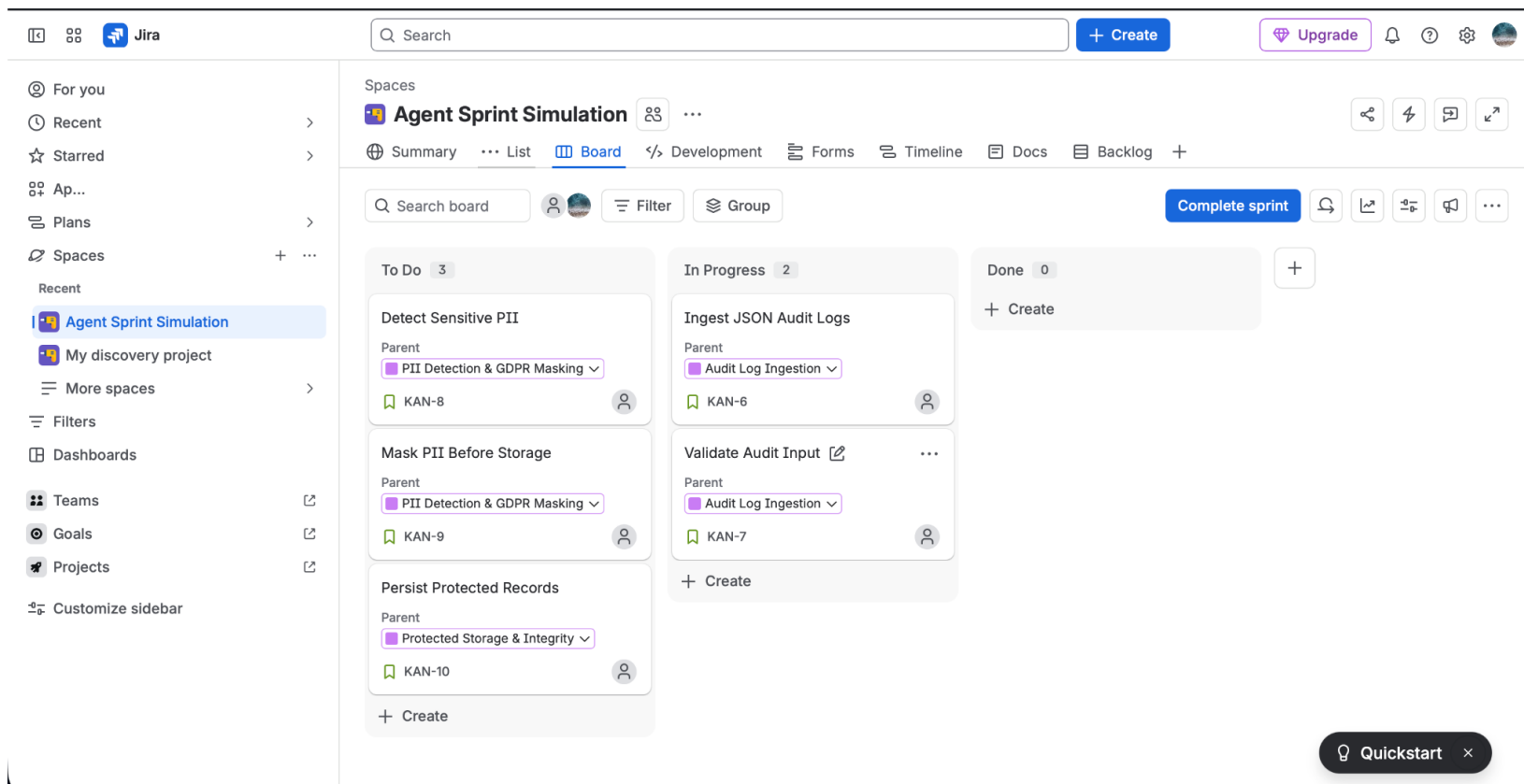


Figure 2. Jira Active Sprint evidence during Sprint 1 progress simulation.

5. Sprint 2 Simulation

Planned duration: 1 week **Stories:** KAN-11 to KAN-15 **Planned capacity:** 29 SP

Issue	Story	SP	Final status
KAN-11	Generate SHA-256 Integrity Hash	5	Done
KAN-12	Query & Filter Audit History	5	Done
KAN-13	Generate Immutable Audit Export	8	Done
KAN-14	Secure Authorized Audit Access	3	Done
KAN-15	Meet Large-Scale Query Target	8	Done

Sprint 2 was started after Sprint 1 was completed. The captured Jira board shows the five Sprint 2 stories initially in To Do.

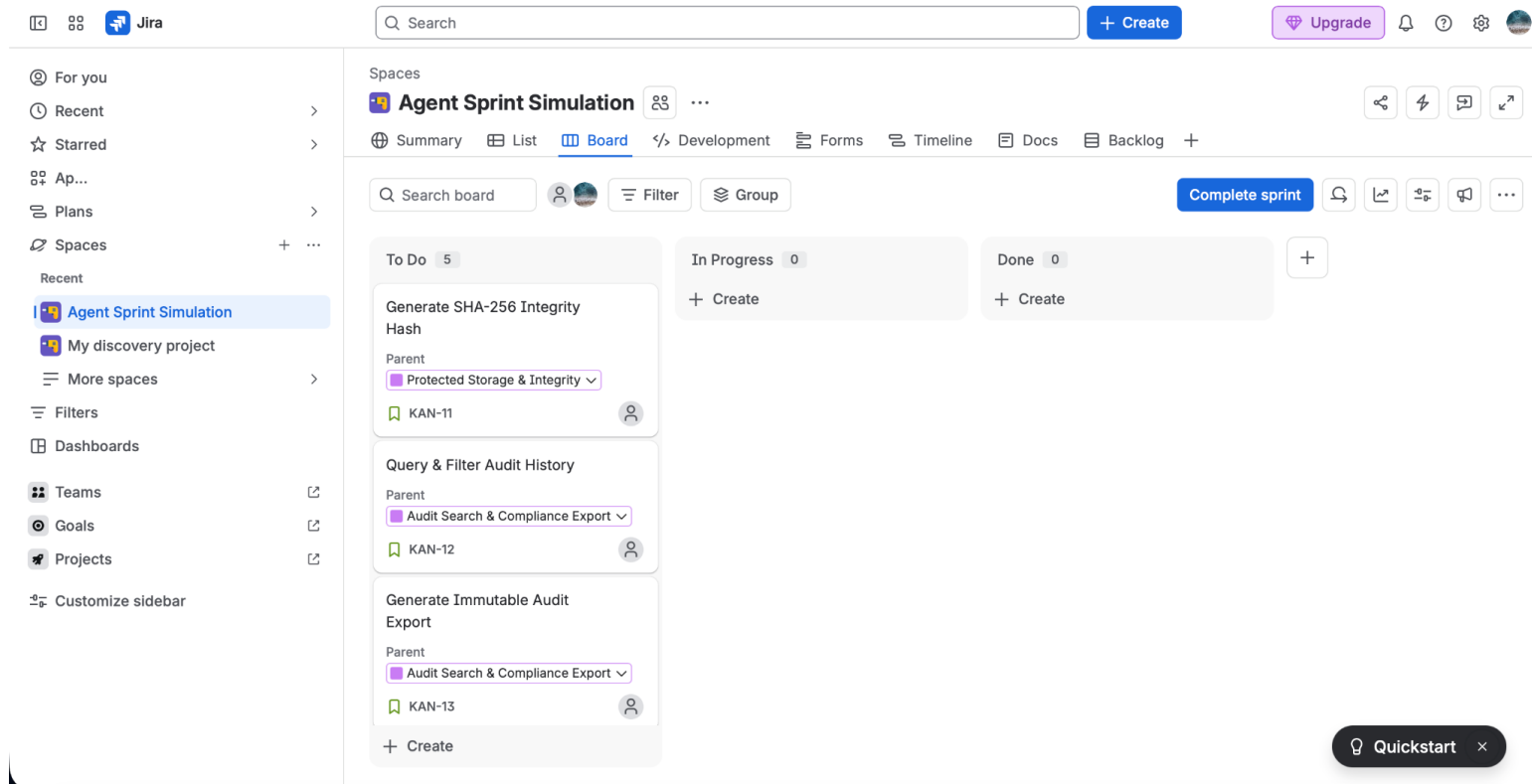


Figure 3. Jira Active Sprint evidence for Sprint 2 at the start of execution.

6. Sprint Completion Evidence

Both one-week sprint simulations were completed. The final Sprint 2 board shows all five Sprint 2 items in Done, with no remaining To Do or In Progress items.

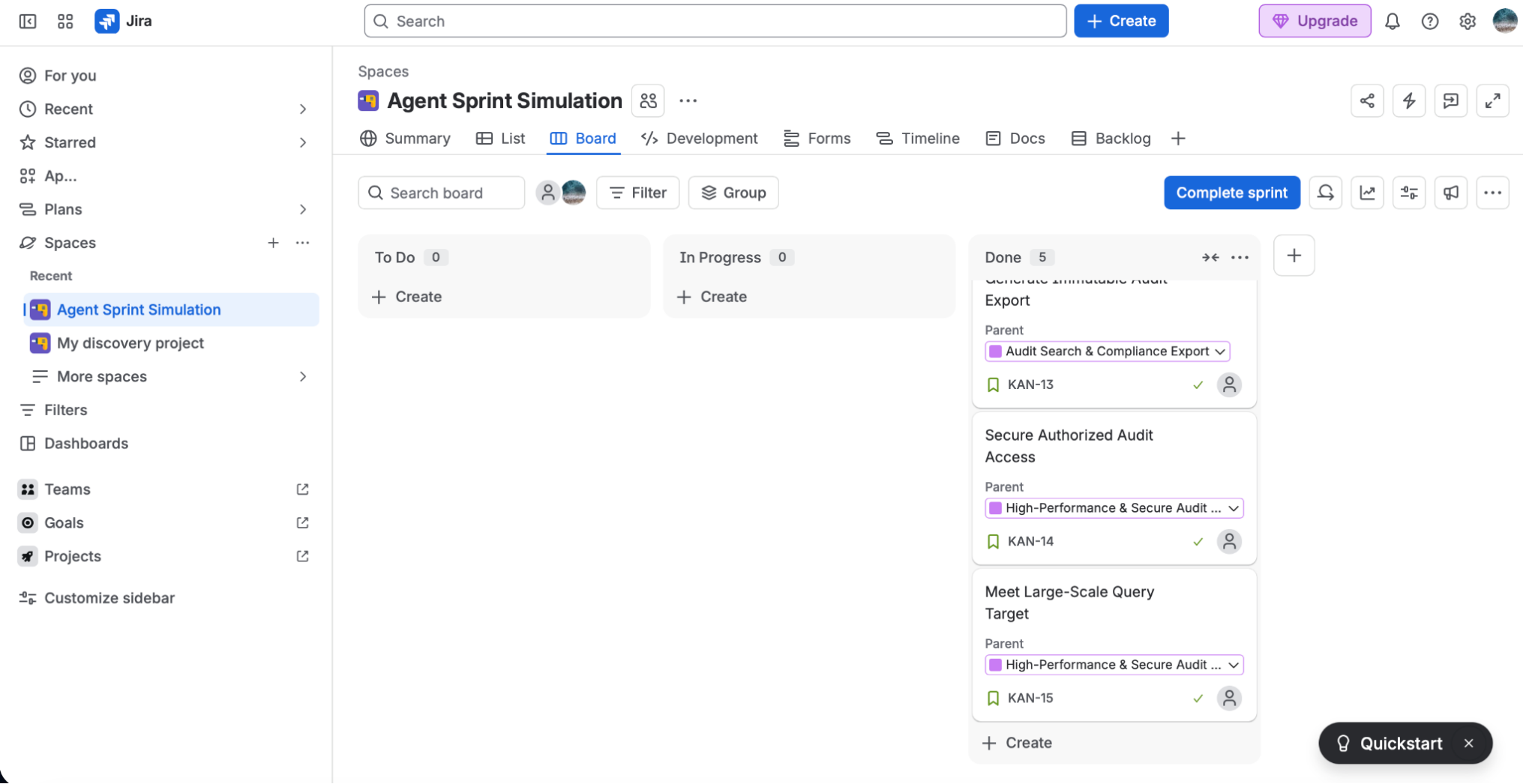


Figure 4. Jira board after Sprint 2 completion: 5 items in Done.

Summary
List
Board
Development
Forms
Timeline
Docs
Backlog

☐
▼
KAN Sprint 1
Add dates
(5 work items)
5
0
0
Start sprint
...

☐	🔖 KAN-6 <u>Ingest JSON Audit Logs</u>	📌 Audit Log Ingestion	To Do ▼	👤 ...
	🔖 KAN-7 Validate Audit Input	📌 Audit Log Ingestion	To Do ▼	👤
	🔖 KAN-8 Detect Sensitive PII	📌 PII Detection & G...	To Do ▼	👤
	🔖 KAN-9 Mask PII Before Storage	📌 PII Detection & G...	To Do ▼	👤
	🔖 KAN-10 Persist Protected Records	📌 Protected Storage ...	To Do ▼	👤

+ Create

☐
▼
Backlog
(5 work items)
5
0
0
0
Create sprint

5 of 5 work items visible

Figure 5. Jira backlog after the two completed sprint cycles; the 10 work items are distributed across the completed sprint history.

7. Burndown Charts

The burndown charts track remaining story points across each seven-day sprint. The planned work reaches zero by the end of each sprint.

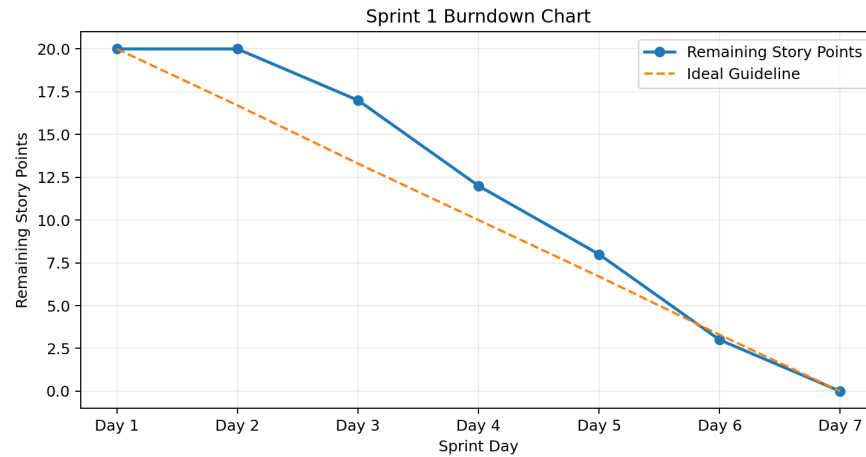


Figure 6. Sprint 1 burndown — 21 SP → 0 SP.

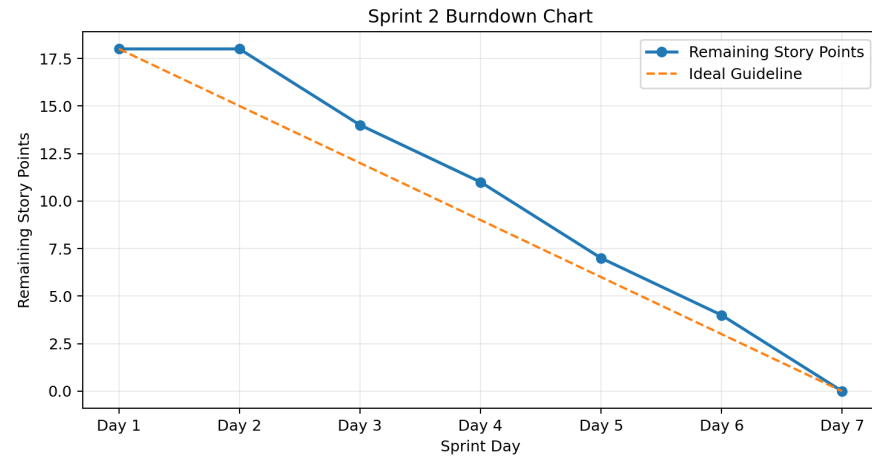


Figure 7. Sprint 2 burndown — 29 SP → 0 SP.

Interpretation: Sprint 1 completed 21 planned story points and Sprint 2 completed 29. The stepwise decline represents work being completed in batches during the simulated sprint rather than continuously.

8. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, at a relative estimation level. The 3-point stories represented comparatively contained work, while 5- and 8-point stories represented greater complexity, integration effort, or uncertainty. The larger estimates for immutable export and large-scale query performance reflect their additional validation, security, and performance risks.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized around dependencies and business/user value. Reliable ingestion and PII protection come first because later audit capabilities depend on protected records. Integrity, search/export, and large-scale secure access follow as higher-level compliance capabilities.

3. How did your simulated sprint align with your plan?

The two simulated sprints followed the planned order and completed all selected stories. Sprint 1 established the ingestion and protection foundation, while Sprint 2 completed integrity, investigation/export, security, and scale-oriented work. The Jira board progression followed To Do → In Progress → Done.

4. What insights did the burndown chart give about your team's capacity?

Both charts reach zero remaining work within the planned one-week sprint. Sprint 2 carried more points than Sprint 1, indicating higher planned capacity but also higher risk because it contains the 8-point export and 8-point performance stories. In a real project, repeated deviation from the ideal guideline would be used to adjust future capacity and sprint planning.